

README: Project 4 – Simple Router

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Test ID	Command	Description
T1	<code>client1 ping 192.168.2.2</code>	Tests basic ARP broadcast/reply exchange and ICMP Echo Reply generation. Confirms that the router responds to echo requests directed to its interface and properly resets TTL.
T2	<code>client1 traceroute 192.168.2.2</code>	Verifies TTL decrement at each hop and correct generation of ICMP Time Exceeded messages from the router during intermediate forwarding.
T3	<code>client1 ping 8.8.8.8</code>	Tests behavior when destination is unreachable due to no matching route in the routing table. Router must respond with ICMP Network Unreachable.
T4	<code>client1 ping 192.168.2.99</code>	Ensures ARP retry logic is triggered and fails after 5 attempts, followed by correct generation of ICMP Host Unreachable.
T5	<code>client1 wget 10.0.1.1</code>	Sends UDP to a closed port on the router to test generation of ICMP Port Unreachable when transport protocol reaches an unsupported endpoint.
T6	<code>client2 ping 10.0.1.100</code>	Tests enforcement of IP blacklist policy for source IPs within 10.0.2.0/24. Packet should be dropped silently with router's logs.
T7	<code>client1 ping 10.0.2.100</code>	Tests enforcement of IP blacklist policy for destination IPs within 10.0.2.0/24. Packet should be dropped silently with router's logs.

Test 1 result

4 5.535576	e6:4c:e3:33:f2:4d	Broadcast	ARP	42 Who has 10.0.1.1? Tell 10.0.1.100
5 5.535655	32:73:4c:b8:ca:f6	e6:4c:e3:33:f2:4d	ARP	42 10.0.1.1 is at 32:73:4c:b8:ca:f6
6 5.543950	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=1/256, ttl=64 (no response found!)
7 5.544508	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.2? Tell 192.168.2.1
8 5.611847	46:d6:d3:7b:b5:2f	be:aa:48:f9:14:96	ARP	42 192.168.2.2 is at 46:d6:d3:7b:b5:2f
9 5.612031	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=1/256, ttl=63 (reply in 8)
10 5.686111	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=1/256, ttl=64 (request in 9)
11 5.686336	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
12 5.755880	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
13 5.755953	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=1/256, ttl=63
14 5.652945	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=2/512, ttl=64 (no response found!)
15 5.652969	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=2/512, ttl=63 (reply in 16)
16 5.673947	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=2/512, ttl=64 (request in 15)
17 5.673975	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=2/512, ttl=63
18 5.641179	fe80::d8d1:24ff:fe...	ff02::2	ICMPv6	70 Router Solicitation from da:d1:24:1c:3d:c4
19 5.555778	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=3/768, ttl=64 (no response found!)
20 5.555872	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=3/768, ttl=63 (reply in 21)
21 5.557736	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=3/768, ttl=64 (request in 20)
22 5.557759	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=3/768, ttl=63
23 5.579327	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=4/1024, ttl=64 (no response found!)
24 5.579395	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=4/1024, ttl=63 (reply in 25)
25 5.581857	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=4/1024, ttl=64 (request in 24)
26 5.581879	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=4/1024, ttl=63
27 5.748321	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=5/1280, ttl=64 (no response found!)
28 5.748347	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=5/1280, ttl=63 (reply in 29)
29 5.762163	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=5/1280, ttl=64 (request in 28)
30 5.762428	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=5/1280, ttl=63
31 10.541347	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=6/1536, ttl=64 (no response found!)
32 10.541371	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=6/1536, ttl=63 (reply in 33)
33 10.544255	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=6/1536, ttl=64 (request in 32)
34 10.544297	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=6/1536, ttl=63
35 10.763181	46:d6:d3:7b:b5:2f	be:aa:48:f9:14:96	ARP	42 Who has 192.168.2.1? Tell 192.168.2.2
36 10.763131	be:aa:48:f9:14:96	46:d6:d3:7b:b5:2f	ARP	42 192.168.2.1 is at be:aa:48:f9:14:96
37 11.565304	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=7/1792, ttl=64 (no response found!)
38 11.565330	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=7/1792, ttl=63 (reply in 39)
39 11.587016	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=7/1792, ttl=64 (request in 38)
40 11.587081	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=7/1792, ttl=63
41 12.583738	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=8/2048, ttl=64 (no response found!)
42 12.583772	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=8/2048, ttl=63 (reply in 43)
43 12.586740	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=8/2048, ttl=64 (request in 42)
44 12.586758	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=8/2048, ttl=63
45 13.597284	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=9/2304, ttl=64 (no response found!)
46 13.597309	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=9/2304, ttl=63 (reply in 47)
47 13.599175	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=9/2304, ttl=64 (request in 46)
48 13.599335	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=9/2304, ttl=63
49 14.596056	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=10/2560, ttl=64 (no response found!)
50 14.596118	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=10/2560, ttl=63 (reply in 51)
51 14.597605	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=10/2560, ttl=64 (request in 50)
52 14.597653	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=10/2560, ttl=63
53 15.667453	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=11/2816, ttl=64 (no response found!)
54 15.667522	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=11/2816, ttl=63 (reply in 55)
55 15.669563	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=11/2816, ttl=64 (request in 54)
56 15.669610	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=11/2816, ttl=63
57 16.647091	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=12/3072, ttl=64 (no response found!)
58 16.647119	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=12/3072, ttl=63 (reply in 59)
59 16.648403	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=12/3072, ttl=64 (request in 58)
60 16.648448	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=12/3072, ttl=63
61 17.609817	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=13/3328, ttl=64 (no response found!)
62 17.609913	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=13/3328, ttl=63 (reply in 63)
63 17.614106	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=13/3328, ttl=64 (request in 62)
64 17.614165	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=13/3328, ttl=63
65 18.635814	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=14/3584, ttl=64 (no response found!)
66 18.635877	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=14/3584, ttl=63 (reply in 67)
67 18.636571	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=14/3584, ttl=64 (request in 66)
68 18.636623	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=14/3584, ttl=63
69 19.666714	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=15/3840, ttl=64 (no response found!)
70 19.666795	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=15/3840, ttl=63 (reply in 71)
71 19.618443	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=15/3840, ttl=64 (request in 70)
72 19.618677	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=15/3840, ttl=63
73 20.723353	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=16/4096, ttl=64 (no response found!)
74 20.723501	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=16/4096, ttl=63 (reply in 75)
75 20.728565	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=16/4096, ttl=64 (request in 74)
76 20.728617	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=16/4096, ttl=63
77 21.685978	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=17/4352, ttl=64 (no response found!)
78 21.686065	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.2? Tell 192.168.2.1
79 21.687718	46:d6:d3:7b:b5:2f	be:aa:48:f9:14:96	ARP	42 192.168.2.2 is at 46:d6:d3:7b:b5:2f
80 21.687760	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=17/4352, ttl=63 (reply in 81)
81 21.776466	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=17/4352, ttl=64 (request in 80)
82 21.776557	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
83 21.849312	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
84 21.849367	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=17/4352, ttl=63
85 22.659289	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=18/4608, ttl=64 (no response found!)
86 22.659374	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=18/4608, ttl=63 (reply in 87)
87 22.661638	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=18/4608, ttl=64 (request in 86)
88 22.661683	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=18/4608, ttl=63
89 23.659955	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=19/4864, ttl=64 (no response found!)
90 23.660001	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=19/4864, ttl=63 (reply in 91)
91 23.661627	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=19/4864, ttl=64 (request in 90)
92 23.661688	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=19/4864, ttl=63
93 24.649478	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=20/5120, ttl=64 (no response found!)
94 24.649593	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=20/5120, ttl=63 (reply in 95)
95 24.643869	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=20/5120, ttl=64 (request in 94)
96 24.643889	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=20/5120, ttl=63
97 25.649149	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=21/5376, ttl=64 (no response found!)
98 25.649172	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=21/5376, ttl=63 (reply in 99)
99 25.657681	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=21/5376, ttl=64 (request in 98)
100 25.658029	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=21/5376, ttl=63
101 26.675038	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=22/5632, ttl=64 (no response found!)
102 26.675117	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=22/5632, ttl=63 (reply in 103)
103 26.680342	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=22/5632, ttl=64 (request in 102)
104 26.680369	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=22/5632, ttl=63
105 27.636885	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=23/5888, ttl=64 (no response found!)
106 27.636938	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=23/5888, ttl=63 (reply in 107)
107 27.642428	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=23/5888, ttl=64 (request in 106)
108 27.642479	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=23/5888, ttl=63
109 28.677540	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=24/6144, ttl=64 (no response found!)
110 28.677599	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=24/6144, ttl=63 (reply in 111)
111 28.680863	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=24/6144, ttl=64 (request in 110)
112 28.680912	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=24/6144, ttl=63
113 29.181974	fe80::78e7:65ff:fe8...	ff02::2	ICMPv6	70 Router Solicitation from 7a:e7:65:87:e3:d9
114 29.606089	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=25/6400, ttl=64 (no response found!)
115 29.606712	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=25/6400, ttl=63 (reply in 116)
116 29.664260	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=25/6400, ttl=64 (request in 115)
117 29.664275	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=25/6400, ttl=63
118 30.659503	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=26/6656, ttl=64 (no response found!)
119 30.659559	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=26/6656, ttl=63 (reply in 120)
120 30.661463	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=26/6656, ttl=64 (request in 119)
121 30.661475	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=26/6656, ttl=63
122 31.247972	fe80::e44c:e3ff:fe3...	ff02::2	ICMPv6	70 Router Solicitation from e6:4c:e3:33:f2:4d
123 31.674472	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=27/6912, ttl=64 (no response found!)
124 31.674502	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=27/6912, ttl=63 (reply in 125)
125 31.684699	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=27/6912, ttl=64 (request in 124)
126 31.684720	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=27/6912, ttl=63
127 32.841771	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=28/7168, ttl=64 (no response found!)
128 32.841817	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request id=0x3243, seq=28/7168, ttl=63 (reply in 129)
129 32.844805	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply id=0x3243, seq=28/7168, ttl=64 (request in 128)
130 32.844817	192.168.2.2	1		

135	34.666317	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request	id=0x3243, seq=30/7680, ttl=64 (no response found!)
136	34.666344	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request	id=0x3243, seq=30/7680, ttl=63 (reply in 137)
137	34.668995	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply	id=0x3243, seq=30/7680, ttl=64 (request in 136)
138	34.669078	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply	id=0x3243, seq=30/7680, ttl=63
139	35.360359	fe80::44d6:d3ff:fe7... ff02::2		ICMPv6	70 Router Solicitation	from 46:d6:d3:7b:b5:2f
140	35.695128	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request	id=0x3243, seq=31/7936, ttl=64 (no response found!)
141	35.695155	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request	id=0x3243, seq=31/7936, ttl=63 (reply in 142)
142	35.698233	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply	id=0x3243, seq=31/7936, ttl=64 (request in 141)
143	35.698264	192.168.2.2	10.0.1.100	ICMP	98 Echo (ping) reply	id=0x3243, seq=31/7936, ttl=63
144	36.706099	10.0.1.100	192.168.2.2	ICMP	98 Echo (ping) request	id=0x3243, seq=32/8192, ttl=64 (no response found!)

```
mininet> client1 ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
64 bytes from 192.168.2.2: icmp_seq=1 ttl=63 time=287 ms
64 bytes from 192.168.2.2: icmp_seq=2 ttl=63 time=87.8 ms
64 bytes from 192.168.2.2: icmp_seq=3 ttl=63 time=76.4 ms
64 bytes from 192.168.2.2: icmp_seq=4 ttl=63 time=97.1 ms
64 bytes from 192.168.2.2: icmp_seq=5 ttl=63 time=407 ms
64 bytes from 192.168.2.2: icmp_seq=6 ttl=63 time=62.5 ms
64 bytes from 192.168.2.2: icmp_seq=7 ttl=63 time=102 ms
64 bytes from 192.168.2.2: icmp_seq=8 ttl=63 time=75.9 ms
64 bytes from 192.168.2.2: icmp_seq=9 ttl=63 time=86.3 ms
64 bytes from 192.168.2.2: icmp_seq=10 ttl=63 time=82.3 ms
64 bytes from 192.168.2.2: icmp_seq=11 ttl=63 time=119 ms
64 bytes from 192.168.2.2: icmp_seq=12 ttl=63 time=95.2 ms
64 bytes from 192.168.2.2: icmp_seq=13 ttl=63 time=57.6 ms
64 bytes from 192.168.2.2: icmp_seq=14 ttl=63 time=81.5 ms
64 bytes from 192.168.2.2: icmp_seq=15 ttl=63 time=50.1 ms
64 bytes from 192.168.2.2: icmp_seq=16 ttl=63 time=283 ms
64 bytes from 192.168.2.2: icmp_seq=17 ttl=63 time=276 ms
64 bytes from 192.168.2.2: icmp_seq=18 ttl=63 time=90.3 ms
64 bytes from 192.168.2.2: icmp_seq=19 ttl=63 time=87.8 ms
64 bytes from 192.168.2.2: icmp_seq=20 ttl=63 time=65.5 ms
64 bytes from 192.168.2.2: icmp_seq=21 ttl=63 time=74.6 ms
64 bytes from 192.168.2.2: icmp_seq=22 ttl=63 time=97.1 ms
64 bytes from 192.168.2.2: icmp_seq=23 ttl=63 time=58.1 ms
64 bytes from 192.168.2.2: icmp_seq=24 ttl=63 time=99.0 ms
64 bytes from 192.168.2.2: icmp_seq=25 ttl=63 time=76.9 ms
64 bytes from 192.168.2.2: icmp_seq=26 ttl=63 time=64.3 ms
64 bytes from 192.168.2.2: icmp_seq=27 ttl=63 time=95.4 ms
64 bytes from 192.168.2.2: icmp_seq=28 ttl=63 time=246 ms
64 bytes from 192.168.2.2: icmp_seq=29 ttl=63 time=96.6 ms
64 bytes from 192.168.2.2: icmp_seq=30 ttl=63 time=66.9 ms
64 bytes from 192.168.2.2: icmp_seq=31 ttl=63 time=90.3 ms
64 bytes from 192.168.2.2: icmp_seq=32 ttl=63 time=101 ms
64 bytes from 192.168.2.2: icmp_seq=33 ttl=63 time=221 ms
64 bytes from 192.168.2.2: icmp_seq=34 ttl=63 time=58.0 ms
64 bytes from 192.168.2.2: icmp_seq=35 ttl=63 time=56.2 ms
64 bytes from 192.168.2.2: icmp_seq=36 ttl=63 time=47.1 ms
64 bytes from 192.168.2.2: icmp_seq=37 ttl=63 time=90.8 ms
^C
```

Test 2 result


```
mininet> client1 traceroute 192.168.2.2
traceroute to 192.168.2.2 (192.168.2.2), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 242.366 ms 242.156 ms 241.735 ms
 2 * * *
 3 * * *
 4 * 192.168.2.2 (192.168.2.2) 335.658 ms 335.631 ms
mininet> client1 traceroute 192.168.2.2
traceroute to 192.168.2.2 (192.168.2.2), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 46.184 ms 46.142 ms 50.261 ms
 2 192.168.2.2 (192.168.2.2) 128.211 ms 128.343 ms 153.416 ms
mininet>
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.1.100	192.168.2.2	UDP	74	38904 → 33434 Len=32
2	0.000140	32:73:4c:b8:ca:f6	Broadcast	ARP	42	Who has 10.0.1.100? Tell 10.0.1.1
3	0.033183	10.0.1.100	192.168.2.2	UDP	74	45758 → 33435 Len=32
4	0.075797	10.0.1.100	192.168.2.2	UDP	74	44623 → 33436 Len=32
5	0.075830	10.0.1.100	192.168.2.2	UDP	74	49526 → 33437 Len=32
6	0.075834	be:aa:48:f9:14:96	Broadcast	ARP	42	Who has 192.168.2.2? Tell 192.168.2.1
7	0.075887	10.0.1.100	192.168.2.2	UDP	74	44186 → 33438 Len=32
8	0.075892	10.0.1.100	192.168.2.2	UDP	74	36261 → 33439 Len=32
9	0.075894	10.0.1.100	192.168.2.2	UDP	74	43396 → 33440 Len=32
10	0.075896	10.0.1.100	192.168.2.2	UDP	74	46134 → 33441 Len=32
11	0.075898	10.0.1.100	192.168.2.2	UDP	74	48312 → 33442 Len=32
12	0.075900	10.0.1.100	192.168.2.2	UDP	74	41742 → 33443 Len=32
13	0.075901	10.0.1.100	192.168.2.2	UDP	74	35570 → 33444 Len=32
14	0.075903	10.0.1.100	192.168.2.2	UDP	74	56836 → 33445 Len=32
15	0.075905	10.0.1.100	192.168.2.2	UDP	74	43343 → 33446 Len=32
16	0.075910	10.0.1.100	192.168.2.2	UDP	74	43326 → 33447 Len=32
17	0.075912	10.0.1.100	192.168.2.2	UDP	74	37951 → 33448 Len=32
18	0.075913	10.0.1.100	192.168.2.2	UDP	74	55503 → 33449 Len=32
19	0.075916	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42	10.0.1.100 is at e6:4c:e3:33:f2:4d
20	0.075917	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
21	0.075919	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
22	0.075920	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
23	0.144102	46:d6:d3:7b:b5:2f	be:aa:48:f9:14:96	ARP	42	192.168.2.2 is at 46:d6:d3:7b:b5:2f
24	0.144126	10.0.1.100	192.168.2.2	UDP	74	55503 → 33449 Len=32
25	0.144135	10.0.1.100	192.168.2.2	UDP	74	37951 → 33448 Len=32
26	0.144137	10.0.1.100	192.168.2.2	UDP	74	43326 → 33447 Len=32
27	0.144139	10.0.1.100	192.168.2.2	UDP	74	43343 → 33446 Len=32
28	0.144140	10.0.1.100	192.168.2.2	UDP	74	56836 → 33445 Len=32
29	0.144141	10.0.1.100	192.168.2.2	UDP	74	35570 → 33444 Len=32
30	0.144142	10.0.1.100	192.168.2.2	UDP	74	41742 → 33443 Len=32
31	0.144143	10.0.1.100	192.168.2.2	UDP	74	48312 → 33442 Len=32
32	0.144144	10.0.1.100	192.168.2.2	UDP	74	46134 → 33441 Len=32
33	0.144145	10.0.1.100	192.168.2.2	UDP	74	43396 → 33440 Len=32
34	0.144146	10.0.1.100	192.168.2.2	UDP	74	36261 → 33439 Len=32
35	0.144147	10.0.1.100	192.168.2.2	UDP	74	44186 → 33438 Len=32
36	0.144148	10.0.1.100	192.168.2.2	UDP	74	49526 → 33437 Len=32
37	0.227537	10.0.1.100	192.168.2.2	UDP	74	49462 → 33450 Len=32
38	0.227562	10.0.1.100	192.168.2.2	UDP	74	49462 → 33450 Len=32
39	0.227620	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
40	0.227624	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
41	0.227629	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
42	0.227630	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
43	0.229297	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
44	0.229311	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
45	0.229316	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
46	0.229327	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
47	0.230756	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
48	0.230769	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
49	0.230775	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
50	0.230777	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
51	0.230779	10.0.1.100	192.168.2.2	UDP	74	33438 → 33451 Len=32
52	0.230780	10.0.1.100	192.168.2.2	UDP	74	33438 → 33451 Len=32
53	0.230782	10.0.1.100	192.168.2.2	UDP	74	60577 → 33452 Len=32
54	0.230784	10.0.1.100	192.168.2.2	UDP	74	60577 → 33452 Len=32
55	4.624580	10.0.1.100	192.168.2.2	UDP	74	58161 → 33434 Len=32
56	4.626407	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
57	4.626819	10.0.1.100	192.168.2.2	UDP	74	49374 → 33435 Len=32
58	4.626843	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
59	4.626856	10.0.1.100	192.168.2.2	UDP	74	46409 → 33436 Len=32
60	4.626859	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
61	4.626866	10.0.1.100	192.168.2.2	UDP	74	44004 → 33437 Len=32
62	4.626869	10.0.1.100	192.168.2.2	UDP	74	44004 → 33437 Len=32
63	4.637950	10.0.1.100	192.168.2.2	UDP	74	43883 → 33438 Len=32
64	4.637975	10.0.1.100	192.168.2.2	UDP	74	43883 → 33438 Len=32
65	4.643313	10.0.1.100	192.168.2.2	UDP	74	48658 → 33439 Len=32
66	4.643335	10.0.1.100	192.168.2.2	UDP	74	48658 → 33439 Len=32
67	4.647467	10.0.1.100	192.168.2.2	UDP	74	54202 → 33440 Len=32
68	4.647490	10.0.1.100	192.168.2.2	UDP	74	54202 → 33440 Len=32
69	4.649373	10.0.1.100	192.168.2.2	UDP	74	43052 → 33441 Len=32
70	4.649391	10.0.1.100	192.168.2.2	UDP	74	43052 → 33441 Len=32
71	4.650152	10.0.1.100	192.168.2.2	UDP	74	59145 → 33442 Len=32
72	4.650161	10.0.1.100	192.168.2.2	UDP	74	59145 → 33442 Len=32
73	4.660222	10.0.1.100	192.168.2.2	UDP	74	55005 → 33443 Len=32
74	4.660243	10.0.1.100	192.168.2.2	UDP	74	55005 → 33443 Len=32
75	4.660248	10.0.1.100	192.168.2.2	UDP	74	33420 → 33444 Len=32
76	4.660249	10.0.1.100	192.168.2.2	UDP	74	33420 → 33444 Len=32
77	4.661127	10.0.1.100	192.168.2.2	UDP	74	41875 → 33445 Len=32
78	4.661135	10.0.1.100	192.168.2.2	UDP	74	41875 → 33445 Len=32
79	4.664406	10.0.1.100	192.168.2.2	UDP	74	37452 → 33446 Len=32
80	4.664425	10.0.1.100	192.168.2.2	UDP	74	37452 → 33446 Len=32
81	4.664429	10.0.1.100	192.168.2.2	UDP	74	50642 → 33447 Len=32
82	4.664430	10.0.1.100	192.168.2.2	UDP	74	50642 → 33447 Len=32
83	4.664432	10.0.1.100	192.168.2.2	UDP	74	41466 → 33448 Len=32
84	4.664434	10.0.1.100	192.168.2.2	UDP	74	41466 → 33448 Len=32
85	4.665008	10.0.1.100	192.168.2.2	UDP	74	46584 → 33449 Len=32
86	4.665106	10.0.1.100	192.168.2.2	UDP	74	46584 → 33449 Len=32
87	4.672504	10.0.1.100	192.168.2.2	UDP	74	38990 → 33450 Len=32
88	4.672552	10.0.1.100	192.168.2.2	UDP	74	38990 → 33450 Len=32
89	4.672567	10.0.1.100	192.168.2.2	UDP	74	33909 → 33451 Len=32
90	4.672587	10.0.1.100	192.168.2.2	UDP	74	33909 → 33451 Len=32
91	4.672592	10.0.1.100	192.168.2.2	UDP	74	60453 → 33452 Len=32
92	4.672594	10.0.1.100	192.168.2.2	UDP	74	60453 → 33452 Len=32
93	4.681751	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
94	4.681775	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
95	4.683569	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
96	4.683587	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
97	4.690185	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
98	4.690212	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
99	4.690219	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
100	4.690220	192.168.2.2	10.0.1.100	ICMP	102	Destination unreachable (Port unreachable)
101	4.972732	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42	Who has 10.0.1.1? Tell 10.0.1.100
102	4.972754	32:73:4c:b8:ca:f6	e6:4c:e3:33:f2:4d	ARP	42	10.0.1.1 is at 32:73:4c:b8:ca:f6

Test 3 result

```
mininet> client1 ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
From 10.0.1.1 icmp_seq=1 Destination Net Unreachable
From 10.0.1.1 icmp_seq=2 Destination Net Unreachable
From 10.0.1.1 icmp_seq=3 Destination Net Unreachable
From 10.0.1.1 icmp_seq=4 Destination Net Unreachable
From 10.0.1.1 icmp_seq=5 Destination Net Unreachable
From 10.0.1.1 icmp_seq=6 Destination Net Unreachable
^C
--- 8.8.8.8 ping statistics ---
6 packets transmitted, 0 received, +6 errors, 100% packet loss, time 5007ms
```

1	0.000000	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=1/256, ttl=64 (no response found!)
2	0.000126	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
3	0.004234	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
4	0.004825	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
5	1.005464	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=2/512, ttl=64 (no response found!)
6	1.005512	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
7	2.033182	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=3/768, ttl=64 (no response found!)
8	2.033231	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
9	3.030822	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=4/1024, ttl=64 (no response found!)
10	3.030877	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
11	4.007195	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=5/1280, ttl=64 (no response found!)
12	4.007246	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
13	4.993299	10.0.1.100	8.8.8.8	ICMP	98 Echo (ping) request id=0xe888, seq=6/1536, ttl=64 (no response found!)
14	4.993372	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Network unreachable)
15	5.276617	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 Who has 10.0.1.1? Tell 10.0.1.100
16	5.276642	32:73:4c:b8:ca:f6	e6:4c:e3:33:f2:4d	ARP	42 10.0.1.1 is at 32:73:4c:b8:ca:f6
17	16.496983	fe80::e44c:e3ff:fe3... ff02::2		ICMPv6	70 Router Solicitation from e6:4c:e3:33:f2:4d

Test 4 result

```
mininet> client1 ping 192.168.2.99
PING 192.168.2.99 (192.168.2.99) 56(84) bytes of data.
From 10.0.1.1 icmp_seq=1 Destination Host Unreachable
From 10.0.1.1 icmp_seq=2 Destination Host Unreachable
From 10.0.1.1 icmp_seq=3 Destination Host Unreachable
From 10.0.1.1 icmp_seq=4 Destination Host Unreachable
From 10.0.1.1 icmp_seq=5 Destination Host Unreachable
From 10.0.1.1 icmp_seq=10 Destination Host Unreachable
From 10.0.1.1 icmp_seq=9 Destination Host Unreachable
From 10.0.1.1 icmp_seq=8 Destination Host Unreachable
From 10.0.1.1 icmp_seq=7 Destination Host Unreachable
From 10.0.1.1 icmp_seq=6 Destination Host Unreachable
From 10.0.1.1 icmp_seq=16 Destination Host Unreachable
From 10.0.1.1 icmp_seq=15 Destination Host Unreachable
From 10.0.1.1 icmp_seq=14 Destination Host Unreachable
From 10.0.1.1 icmp_seq=13 Destination Host Unreachable
From 10.0.1.1 icmp_seq=12 Destination Host Unreachable
From 10.0.1.1 icmp_seq=11 Destination Host Unreachable
^C
--- 192.168.2.99 ping statistics ---
17 packets transmitted, 0 received, +16 errors, 100% packet loss, time 16271ms
```

1	0.000000	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=1/256, ttl=64 (no response found!)
2	0.000044	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
3	0.368694	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
4	0.988161	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=2/512, ttl=64 (no response found!)
5	1.375326	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
6	2.011382	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=3/768, ttl=64 (no response found!)
7	2.377484	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
8	3.081822	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=4/1024, ttl=64 (no response found!)
9	3.386110	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
10	4.081095	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=5/1280, ttl=64 (no response found!)
11	4.386407	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
12	4.404423	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
13	4.404490	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
14	4.404534	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
15	4.404538	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
16	4.404539	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
17	4.404540	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
18	5.082847	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=6/1536, ttl=64 (no response found!)
19	5.082951	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
20	5.152309	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 Who has 10.0.1.1? Tell 10.0.1.100
21	5.152372	32:73:4c:b8:ca:f6	e6:4c:e3:33:f2:4d	ARP	42 10.0.1.1 is at 32:73:4c:b8:ca:f6
22	5.387198	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
23	6.116837	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=7/1792, ttl=64 (no response found!)
24	6.389120	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
25	7.109930	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=8/2048, ttl=64 (no response found!)
26	7.390275	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
27	8.131341	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=9/2304, ttl=64 (no response found!)
28	8.392020	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
29	9.170230	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=10/2560, ttl=64 (no response found!)
30	9.170258	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
31	9.170323	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
32	9.170332	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
33	9.170335	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
34	9.170338	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
35	10.189477	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=11/2816, ttl=64 (no response found!)
36	10.189508	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
37	11.171634	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=12/3072, ttl=64 (no response found!)
38	11.171668	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
39	12.182667	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=13/3328, ttl=64 (no response found!)
40	12.182718	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
41	13.217104	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=14/3584, ttl=64 (no response found!)
42	13.217174	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
43	14.263342	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=15/3840, ttl=64 (no response found!)
44	14.263373	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
45	15.269943	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=16/4096, ttl=64 (no response found!)
46	15.270007	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
47	15.270028	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
48	15.270033	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
49	15.270037	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
50	15.270052	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
51	15.270055	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)
52	16.285679	10.0.1.100	192.168.2.99	ICMP	98 Echo (ping) request id=0x89a5, seq=17/4352, ttl=64 (no response found!)
53	16.285727	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
54	17.398443	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
55	18.399149	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
56	19.399742	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
57	20.401292	be:aa:48:f9:14:96	Broadcast	ARP	42 Who has 192.168.2.99? Tell 192.168.2.1
58	21.402892	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
59	21.443281	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
60	21.443307	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Host unreachable)

Test 5 result

```
mininet> client1 wget 10.0.1.1
--2025-05-30 13:38:52-- http://10.0.1.1/
Connecting to 10.0.1.1:80... failed: Connection refused.
```

1	0.000000	10.0.1.100	10.0.1.1	TCP	74 52/710 -- 80 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM=1 TSval=2758246957 TSecr=0 WS=512
2	0.000078	32:73:4c:b8:ca:f6	Broadcast	ARP	42 Who has 10.0.1.100? Tell 10.0.1.1
3	0.000688	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 10.0.1.100 is at e6:4c:e3:33:f2:4d
4	0.009717	10.0.1.1	10.0.1.100	ICMP	70 Destination unreachable (Port unreachable)
5	5.117593	e6:4c:e3:33:f2:4d	32:73:4c:b8:ca:f6	ARP	42 Who has 10.0.1.1? Tell 10.0.1.100
6	5.117621	32:73:4c:b8:ca:f6	e6:4c:e3:33:f2:4d	ARP	42 10.0.1.1 is at 32:73:4c:b8:ca:f6

Test 6 result

```
student@SimpleRouter:~/SimpleRouter/router$ ./sr -l test.pcap
Using VNS sr stub code revised 2009-10-14 (rev 0.20)
Loading routing table from server, clear local routing table.
Loading routing table
```

```
Client student connecting to Server localhost:8888
Requesting topology 0
successfully authenticated as student
Loading routing table from server, clear local routing table.
Loading routing table
```

Router interfaces:

```
<-- Ready to process packets -->
```

7 / 10

[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
^C

1 0.000000	7a:e7:65:87:e3:d9	Broadcast	ARP	42 Who has 10.0.2.1? Tell 10.0.2.100
2 0.000260	72:0a:1a:69:87:a5	7a:e7:65:87:e3:d9	ARP	42 10.0.2.1 is at 72:0a:1a:69:87:a5
3 0.000978	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=1/256, ttl=64 (no response found!)
4 1.036602	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=2/512, ttl=64 (no response found!)
5 2.082661	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=3/768, ttl=64 (no response found!)
6 3.055372	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=4/1024, ttl=64 (no response found!)
7 4.004106	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=5/1280, ttl=64 (no response found!)
8 5.162396	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=6/1536, ttl=64 (no response found!)
9 6.130666	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=7/1792, ttl=64 (no response found!)
10 7.199197	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=8/2048, ttl=64 (no response found!)
11 8.203227	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=9/2304, ttl=64 (no response found!)
12 9.234440	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=10/2560, ttl=64 (no response found!)
13 10.271469	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=11/2816, ttl=64 (no response found!)
14 11.293712	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=12/3072, ttl=64 (no response found!)
15 12.284573	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=13/3328, ttl=64 (no response found!)
16 13.297648	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=14/3584, ttl=64 (no response found!)
17 14.364045	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=15/3840, ttl=64 (no response found!)
18 15.395273	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=16/4096, ttl=64 (no response found!)
19 16.412799	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=17/4352, ttl=64 (no response found!)
20 17.457755	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=18/4608, ttl=64 (no response found!)
21 18.415684	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=19/4864, ttl=64 (no response found!)
22 19.401009	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=20/5120, ttl=64 (no response found!)
23 20.482799	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=21/5376, ttl=64 (no response found!)
24 21.404270	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=22/5632, ttl=64 (no response found!)
25 21.610185	7a:e7:65:87:e3:d9	72:0a:1a:69:87:a5	ARP	42 Who has 10.0.2.1? Tell 10.0.2.100
26 21.610212	72:0a:1a:69:87:a5	7a:e7:65:87:e3:d9	ARP	42 10.0.2.1 is at 72:0a:1a:69:87:a5
27 22.520360	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=23/5888, ttl=64 (no response found!)
28 23.554995	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=24/6144, ttl=64 (no response found!)
29 24.561059	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=25/6400, ttl=64 (no response found!)
30 25.603301	10.0.2.100	10.0.1.100	ICMP	98 Echo (ping) request id=0x8723, seq=26/6656, ttl=64 (no response found!)

Test 7 result


```
mininet> client1 ping 10.0.2.100
PING 10.0.2.100 (10.0.2.100) 56(84) bytes of data.
^C
--- 10.0.2.100 ping statistics ---
17 packets transmitted, 0 received, 100% packet loss, time 16372ms
```

```
student@SimpleRouter:~/SimpleRouter/router$ ./sr -l test.pcap
Using VNS sr stub code revised 2009-10-14 (rev 0.20)
Loading routing table from server, clear local routing table.
Loading routing table
-----
Destination      Gateway          Mask            Iface
10.0.1.100       10.0.1.100      255.255.255.0   eth3
10.0.2.100       10.0.2.100      255.255.255.0   eth4
192.168.2.2      192.168.2.2     255.255.255.0   eth1
172.64.3.10      172.64.3.10     255.255.255.0   eth2
-----
Client student connecting to Server localhost:8888
Requesting topology 0
successfully authenticated as student
Loading routing table from server, clear local routing table.
Loading routing table
-----
Destination      Gateway          Mask            Iface
10.0.1.100       10.0.1.100      255.255.255.0   eth3
10.0.2.100       10.0.2.100      255.255.255.0   eth4
192.168.2.2      192.168.2.2     255.255.255.0   eth1
172.64.3.10      172.64.3.10     255.255.255.0   eth2
-----
Router interfaces:
eth4    MAC Addr: 72:0a:1a:69:87:a5
        IP Addr: 10.0.2.1
eth3    MAC Addr: 32:73:4c:b8:ca:f6
        IP Addr: 10.0.1.1
eth2    MAC Addr: f2:95:e3:c8:6c:75
        IP Addr: 172.64.3.1
eth1    MAC Addr: be:aa:48:f9:14:96
        IP Addr: 192.168.2.1
<-- Ready to process packets -->
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
[IP blocked] : 10.0.2.100
^C
```

Initial TTL

66	11.214223	10.0.1.1	10.0.1.100	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
67	11.214238	10.0.1.100	192.168.2.2	UDP	74 48436 - 33435 Len=32
68	11.214241	10.0.1.1	10.0.1.100	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
Frame 66: 70 bytes on wire (560 bits), 70 bytes captured (560 bits)					
Ethernet II, Src: f2:3e:59:0e:99:1d (f2:3e:59:0e:99:1d), Dst: 66:24:05:3a:06:f6 (66:24:05:3a:06:f6)					
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.1.100					
0100 = Version: 4					
... 0101 = Header Length: 20 bytes (5)					
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)					
Total Length: 56					
Identification: 0x0000 (0)					
Flags: 0x00					
...0 0000 0000 0000 = Fragment Offset: 0					
Time to Live: 255					
Protocol: ICMP (1)					
Header Checksum: 0xa560 [validation disabled]					
[Header checksum status: Unverified]					
Source Address: 10.0.1.1					
Destination Address: 10.0.1.100					
Internet Control Message Protocol					